JUSTINE LUIS S. CALAGUAS

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EDUCATION

Holy Angel University, Angeles City, Philippines — July 2023 – Expected 2027
Bachelor of Science in Computer Science, Dean's Lister

PROJECTS

A Real-Time Driver Drowsiness Recognition Detection System — August 2026

- Designed and implemented a three-component offline embedded system (USB camera → laptop → Android alert device) running without internet dependency, using Python, OpenCV, and Google MediaPipe FaceLandmarker (468 landmarks).
- Trained EyeCNN from scratch on 190,490 eye crop images from YawDD and NTHU-DD datasets using PyTorch on Google Colab T4 GPU, achieving 87.3% subject-independent validation accuracy, 86.21% precision, and 5.54% false-alarm rate.
- Implemented video-based GroupShuffleSplit to eliminate data leakage; confirmed 99.7% label accuracy through manual spot-check of 150 samples.

Speech Command Recognition with Reinforcement Learning-Optimized March 2026 **Threshold Tuning**

- Contributed to the coding of the speech command recognition module, including CNN-based Mel spectrogram classification and integration with the Q-learning threshold-tuning agent.
- Assisted in implementing the audio preprocessing pipeline (WAV-to-Mel-spectrogram conversion using librosa) that fed into the CNN classifier.
- Supported development of the modular pipeline connecting the CNN, NLP intent classifier, and RL decision-threshold components.

HAUtogo: A Professor Navigation Application for Angelites — October 2025

- Led quality assurance testing for both mobile front-end and back-end modules as System Tester, verifying login, authentication, professor locator, and schedule viewer functions.
- Ran 15 test cases across authentication, password recovery, location, and schedule functions, logging 4 defects and re-testing each fix until all passed. Assisted in confirming system readiness for demonstration and final evaluation.

The Effect of Sleep Hours on Productivity — October 2025

- Served as researcher on a System Dynamics study modeling the relationship between sleep duration and productivity, focusing on energy, focus, and task-performance feedback loops.
- Helped develop and validate the simulation algorithm across R and Python, later integrated into a web-based interactive tracker (HTML/CSS/JavaScript).
- Assisted in analyzing simulation output from the Sleep Health and Lifestyle Dataset to identify reinforcing and balancing sleep-productivity cycles.

HAUPokemon: Distributed Monster Tracking System — March 2025

- Contributed as EC2 instance developer, assisting in managing and maintaining cloud infrastructure for a distributed monster tracking system.
- Supported deployment and configuration of AWS EC2 instances, ensuring availability and performance across the distributed architecture.
- Collaborated with the full-stack team on version control and system integration using Git and GitHub.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, PHP, HTML, SQL
- **Frameworks & Libraries:** OpenCV, MediaPipe, PyTorch, TFLite, scikit-learn, librosa, SymPy
- **Backend & Databases:** PostgreSQL, MySQL, Supabase

- **Tools & Platforms:** AWS (EC2, Lambda), Git, GitHub, VS Code, Google Colab, Android Studio
- **Core Concepts:** Computer Vision, Machine Learning, CNN, Embedded Systems, Responsive Web Design, CRUD, QA Testing

CERTIFICATIONS

- Legacy Responsive Web Design V8 Certification (freeCodeCamp, Sep 2024)
- Relational Database V8 Certification (freeCodeCamp, Feb 2025)
- AI Fundamentals with IBM SkillsBuild (Cisco, Mar 2026)
- CCNA: Introduction to Networks (Cisco, Apr 2026)